

M002 Week 5 — English Worksheet

Topic: Mini Cube Puzzle · **Course:** Maze Madness · **Time:** about 45 minutes

This week you finish the **mini cube**. You already have starter code that builds part of the cube. Your job is to **keep it going** — add more decisions with conditionals, and leave a way out through the **top**.

Two ways to finish — pick the one that fits you:

- **Challenge:** add **2 more conditionals**, and make **one of them an OR**.
- **Steady:** keep the same pattern the starter uses and **fill the whole cube**.

Either way: the agent must **break a hole in the top** and climb out.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
repeat 5 times:
  place block
  move forward
```

What shape does the agent leave behind? How long is it?


```
if on the edge OR on the top layer:
  place block
otherwise:
  leave empty
```

Only one condition needs to be true. Which spots get a block? Which stay empty?


```
repeat for each layer:
    fill the layer
move up
if at the top:
    break block above
    climb out
```

Describe in plain English what the agent does, from the first layer to climbing out.

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent should place a block on **every edge** of the layer, then move up. Right now it moves up **before** finishing the layer.

```
place block
move up
place block
place block
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent should leave a block when it is on the **bottom layer** **OR** on an **outer wall**. Right now it only fills the bottom layer.

```
if on the bottom layer:
    place block
```

Hint: one condition is missing. Add the wall check with **OR**.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — After the cube is full, the agent should **break the block above** and climb out the top. Right now it tries to climb **before** making the hole, so it gets stuck.

```
move up  
break block above
```

Hint: make the hole first, then move into it.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Finish the Cube 📷

Open this week's world. You will find the **mini cube starter** — some of the cube is already built. Finish it your way:

Challenge path

1. Keep the starter running.
2. Add **2 more conditionals** that decide where blocks go (for example: a different block on corners, or skip a spot to leave a pattern).
3. Make **one** of them use **OR** (for example: **if on a corner OR on the top → place glass**).
4. Make sure the agent **breaks a hole in the top** and climbs out.

Steady path

1. Keep the **same pattern** the starter uses.
2. Repeat it until the **whole cube is filled**.
3. Make sure the agent **breaks a hole in the top** and climbs out.

When you finish, come back here. Send a **photo or video** of the finished cube (and the hole in the top!), then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I looked at the starter code and saw ...

To keep building, I ...

I used a conditional when ...

I made the hole in the top by ...

One tricky moment was when ...

If I had more time, I would ...

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4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while the agent builds the cube. Talk through it like you are teaching someone who has never seen it. Try to use these words: **cube**, **layer**, **conditional**, **OR**, **hole**.

1. Show the starter cube before you run your code.
2. Run your code and watch the cube fill up.
3. Read your conditional out loud and say what makes it run.
4. Show the agent breaking the hole in the top and climbing out.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.